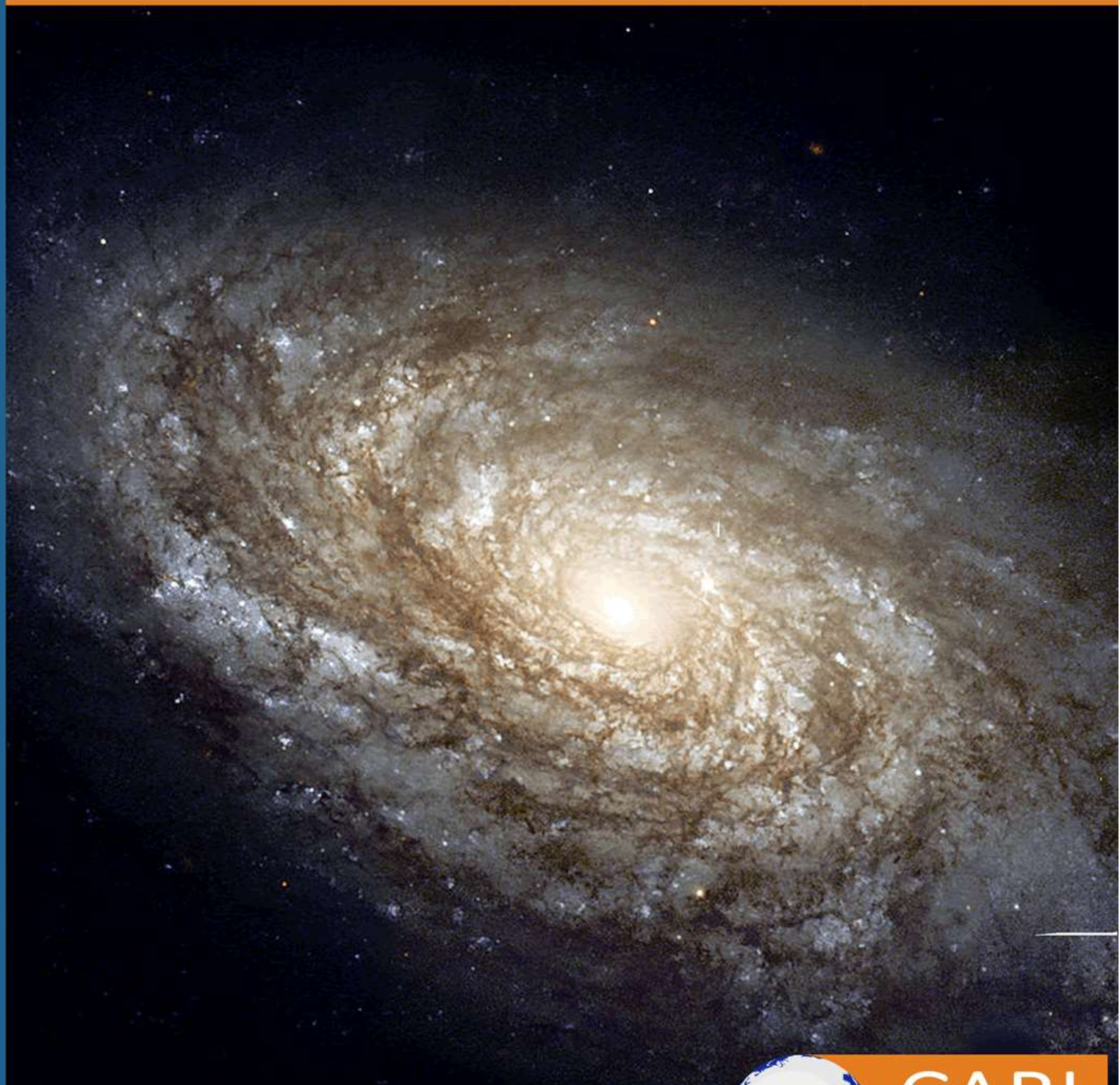


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**Assessment of Fluoride Exposure and Non-Carcinogenic Health
Risks associated with Groundwater Consumption from Olbanita
Boreholes, Nakuru County, Kenya**



Assessment of Fluoride Exposure and Non-Carcinogenic Health Risks associated with Groundwater Consumption from Olbanita Boreholes, Nakuru County, Kenya

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ABSTRACT

Purpose: The aim of this study is therefore to estimate the level of fluoride exposure from groundwater consumption at Olbanita wellfield boreholes and associated potential non-carcinogenic risks to children and adults.

Methodology: Water samples were collected from six productive boreholes (BB2, BB3, BB4, BB5, BB6 and BB7) during dry season of the study area for three months and analyzed for fluoride concentrations using spectrophotometric method. Fluoride exposure levels and potential non carcinogenic health risk were evaluated using the United States Environmental Protection Agency (USEPA) method by calculating Chronic Daily Intake (CDI) and Hazard Quotient (HQ) for adults and children.

Findings: From the results, mean fluoride concentration ranged from 1.50 to 11.60 mg/l, with five of the six boreholes exceeding the World Health Organization (WHO) guideline value of 1.5 mg/l for drinking water. The calculated CDI for adults ranged from 0.0429 to 0.3314 mg/kg/day, while higher values of 0.1000 to 0.7733 mg/kg/day were observed in children population, exceeding USEPA safe oral Reference Dose (RfD) for fluoride of 0.06 mg/kg/day. Corresponding HQ values for adults ranged from 0.71 to 5.52, while children had a recording of 1.67 to 12.89, exceeding the safety threshold of one unit. The findings showed elevated groundwater fluoride contamination in Olbanita boreholes where consumption represents a significant fluoride exposure pathway and potential non-carcinogenic health risk problems for both groups with children being highly vulnerable.

Unique contribution to theory, practice and policy: The study uniquely advances theoretical understanding of the link between groundwater geogenic fluoride enrichment, specific population group exposure and non-carcinogenic health risk assessment using the USEPA CDI and HQ framework. In practice, it provides a risk-based method that can guide identification of high-fluoride groundwater sources and vulnerable consumers to support targeted interventions to reduce exposure. From a policy perspective, the findings can provide evidence base for risk informed groundwater management and public health protection in the study area and other fluoride-prone areas of Kenya's Rift Valley.

Keywords: *Fluoride exposure, groundwater, Non-carcinogenic health risk, Chronic Daily Intake, Hazard quotient, Olbanita boreholes, Nakuru County, Kenya.*

1.0 INTRODUCTION

Groundwater constitutes the primary source of domestic water supply for many rural and peri-urban communities worldwide (WHO, 2022). However, naturally occurring contaminants, among which fluoride is one of the most widespread remains a significant challenge to human health particularly in areas with volcanic and geothermal geological settings (Usman et al., 2024; WHO, 2022; Mwiathi et al., 2022; Ligate et al., 2021; Brindha and Elango, 2011). The association between the active volcanism and dental fluorosis has been described in many volcanic areas worldwide such as Tanzania, Kenya, Ethiopia, Ghana, Nigeria, Malawi, Indonesia and Djibouti (Kitemangu et al., 2026; Usman et al., 2024; Shube et al., 2023; Linhares et al., 2019; Rango et al., 2012; Wambu & Muthakia, 2011).

Globally, more than 200 million people are estimated to consume groundwater containing fluoride concentrations above the World Health Organization guideline value of 1.5 mg/L, with the greatest burden occurring in developing countries (Ali et al., 2026; Kitemangu et al., 2026; Shaji et al., 2024; Podgorski & Berg, 2022; Nordstrom & Smedley, 2022; Tobiloba et al., 2020). In the African context, the East African is identified as one of the world's principal regions affected by naturally elevated groundwater fluoride due to its volcanic geology associated with the East African Rift System (Podgorski & Berg, 2022; Nordstrom & Smedley, 2022; Mwiathi et al., 2022; Ligate et al., 2021; Rango et al., 2012). Consequently, Kenya experiences high fluoride concentrations in groundwater particularly within the Rift valley region, where boreholes and wells remain the major source of domestic water supply (Mwiathi et al., 2022; Gevera & Mouri, 2018). Several studies have documented occurrences and spatial distribution of fluoride in different aquifers, confirming that elevated fluoride remains a significant concern in many parts of Kenya such as Nakuru, Baringo, Laikipia, Kajiado, Narok, Naivasha, Machakos, and Nairobi counties (Keli, 2022; Gevera, 2017; Gevera & Mouri, 2018; Wambu & Muthakia, 2011).

Naturally elevated fluoride presence in groundwater and surface waters can lead to contamination, posing significant non-carcinogenic health risks such as dental and skeletal fluorosis in affected communities, with its primary pathway of exposure being drinking water (Ali et al., 2026; Shaji et al., 2024; Solanki et al., 2022; WHO, 2022; Podgorski & Berg, 2022; Linhares, 2019; Gevera & Mouri, 2018; Fordyce, 2011). Health risk assessment results show that chronic health risks by groundwater geogenic fluoride ingestion are identified to various individuals, with highest threats in children (Ashong et al., 2024; Mwiathi et al., 2022). Chronic exposure primary cause dental and skeletal fluorosis but minimum fluoride intake (0.5 – 1.5 mg/l) by the body helps in bone and dental growth and reduces dental caries (Shaji et al., 2024; WHO, 2022; Tobiloba et al., 2020; Gevera, 2017).

Nakuru County population mainly depends on groundwater for domestic use despite being widely recognized as one of fluoride hotspot regions. Previous studies by Gevera (2017) and Olaka et al. (2016) within Nakuru County consistently reported high occurrences and distribution of fluoride in boreholes often up to 72mg/l and 74.98 mg/l, often exceeding the world health organization

(WHO, 2017) recommended limit of 1.5 mg/l. The authors attributed fluoride enrichment in groundwater–rock interactions within volcanic formations, geothermal activities and evaporation processes. They also observed that in Nakuru groundwater system, aquifers located on the rift floor including Olbanita area, are vulnerable to elevated fluoride concentrations. Consequently, elevated fluoride exposure is an established public-health concern within the wider Nakuru region (Okore et al. 2026; Gevera et al., 2019). Gevera et al. (2019) observed high prevalence of clinically observed dental fluorosis among patients in Gilgil and Njoro related to elevated fluoride exposure. Okore et al. (2026) further documented community experiences of dental discoloration, tooth damage, pain, and social stigma associated with dental fluorosis, demonstrating the potential consequences of prolonged exposure to fluoride-contaminated drinking water.

Although groundwater fluoride enrichment is well documented in the Kenyan Rift Valley, particularly in Nakuru County, existing studies have largely focused on the occurrence, distribution and geological controls of fluoride contamination rather than human health risk assessments. For Olbanita wellfield and its aquifer system, past studies primarily focus on geophysical characterization of lithology and structure, hydrogeochemistry, hydraulic characterization and modeling, groundwater evolutionary processes, spatial distribution of fluoride contamination, seasonal and spatial variation of groundwater quality (Keli, 2022; Sosi et al., 2019), rather than linking fluoride concentration at each borehole to quantitative non-carcinogenic health risks analysis. According to Fawell et al. (2006) fluoride exposure and its associated health effects depend not only on fluoride concentration in drinking water but also on the amount consumed daily, duration of exposure and population traits like age, body weight and climate. Therefore, this study aimed to evaluate fluoride exposure levels and associated specific non-carcinogenic health risks among adults and children from six productive boreholes in Olbanita wellfield, Nakuru County, Kenya.

Specific aim of this study is therefore to; 1. Determine the fluoride concentration in groundwater from six boreholes. 2. Estimate chronic daily fluoride intake among adults and children consuming water from Olbanita boreholes, and 3. Evaluate the potential non-carcinogenic health risks using Hazard Quotient (HQ) approach in reference to USEPA (2011) framework.

Principle novelty of the study is the generation of localized, water source specific evidence linking groundwater contamination fluoride with adult and child quantitative chronic exposure and non- carcinogenic risk in the Olbanita boreholes in Nakuru County Kenya.

2.0 RESEARCH METHODOLOGY

2.1 Study Area Description

The study was conducted at Olbanita wellfield boreholes located in the lower Baringo basin of the Kenyan Rift, Nakuru County, Kenya, approximately 15–20 km northeast of Nakuru City. The boreholes are heavily relied on by Nakuru City and surrounding communities within the Olbanita Corridor for potable water supply. The Olbanita well field is bounded by Menengai Caldera to

the south, Bahati Hills to the east, Solai Escarpment to the north and El Bonwala Hill to the northwest. The Lower Baringo Basin where Olbanita wellfield lies, covers an area of approximately 625 km² and geographically extends between latitude 0° 00' 00''N and 0° 15' 00'' S and longitude 36° 00' 00''E and 36° 15' 00''E. The geology comprises of a volcano-sedimentary system characterized by tertiary to quaternary volcanic formations associated with the evolution of the East African Rift System. The hydrogeology comprises of fractured and weathered volcanic rocks with regional faults enhancing recharge, storage, and flow; and a hydro chemical environment influenced by prolonged groundwater–mineral interactions under alkaline conditions (Sosi et al., 2020; Sosi et al., 2019).

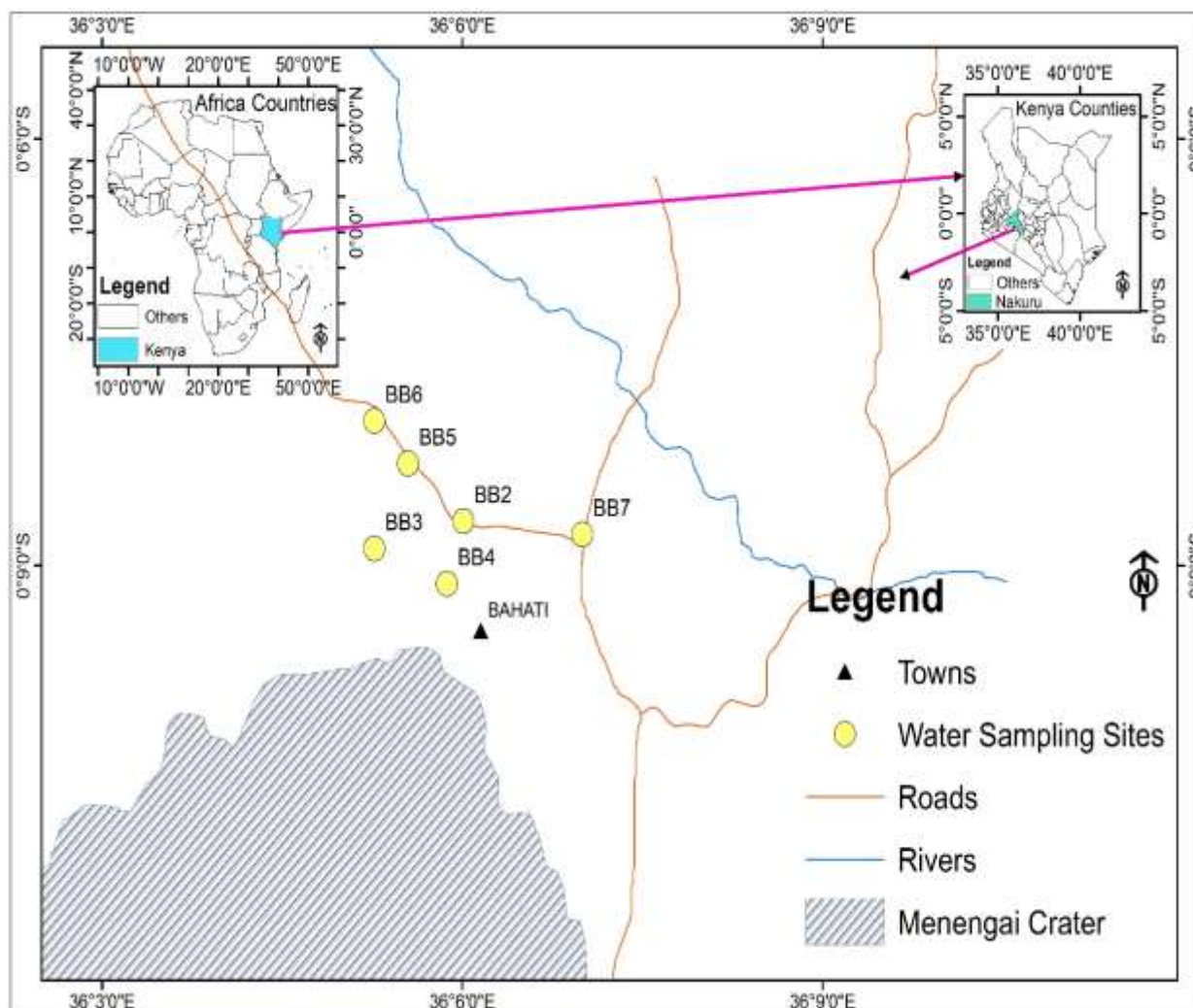


Figure 1: Map showing Location of: (a) Kenya in Africa (b) Nakuru County in Kenya (c) Sampled boreholes (BB2 to BB7) at Olbanita well field, Lower Baringo Basin.

Source: Author

2.2 Sampling design and Data Collection

Representative six productive boreholes (Figure 1) were purposively selected based on accessibility and dependence for water supply. The geographic coordinates of all sampled boreholes were recorded in the field using a hand-held Global Positioning System (GPS) - Garmin etrex-10 GPS where the data were processed using ArcView 10.1 software. Each sampling point was assigned a unique identification code (BB1, BB2, BB3, BB4, BB5, BB6 and BB7) that was used to associate each fluoride concentration with its corresponding borehole location. The measured concentrations were subsequently used as input variables in the fluoride exposure and health-risk calculations. Groundwater samples were collected using standard sampling procedures in clean 250 ml precleaned polyethylene bottles once every month during the dry season of the study area (January to March). Before sampling, boreholes were purged for 3–5 minutes to obtain fresh aquifer water and minimize stagnant water effects. Samples were then labeled with the sampling-point identification and date, preserved, and transported to the laboratory for fluoride analysis.

2.3 Fluoride concentration determination and quality control

Fluoride concentration in groundwater samples was determined using spectrometric method with a UV-Vis spectrometer operated at 570 nm. A standard fluoride solution was used for calibration; whereby measured amounts of the standard and groundwater samples were mixed with measured amounts of water and allowed to stand for the required reaction/development period before their absorbance was measured using the UV-vis spectrometer. The absorbance was then measured at 570nm, and fluoride concentration was determined directly from the spectrometer based on the standard response and reported in mg/l. All samples were analyzed under the same analytical conditions to ensure comparability among the sampling locations.

Quality control was ensured through instrument calibration, standardized analytical procedures, analysis of blanks and duplicate samples and appropriate quality assurance measures during sample analysis to ensure the reliability and accuracy of the fluoride concentration results.

2.4 Non-carcinogenic risk health assessment

The potential non-carcinogenic human health risk associated with fluoride ingestion were evaluated using the United States Environmental Protection Agency (USEPA, 2011) assessment method. Exposure groups considered in the study included children and adults.

2.4.1 Fluoride exposure assessment

Chronic fluoride exposure through drinking-water consumption was evaluated using USEPA Chronic Daily Intake (CDI) approach. CDI estimates the average daily fluoride intake through drinking water (Ashong et al., 2024; USEPA, 2011). CDI was estimated separately for adults and children using the measured fluoride concentration in each borehole water, assumed daily water ingestion rate, exposure frequency, exposure duration, body weight and averaging time (Table 1).

The general relationship was expressed as:

$$CDI = (C \times IR \times EF \times ED) / (BW \times AT)$$

Where; CDI IS Chronic Daily Intake (mg/kg/day)

C is fluoride concentration (mg/l), IR is ingestion rate (l/day), EF is exposure frequency (days/year), ED is exposure duration (Years), BW is body weight (kg), and AT is averaging time (days) where for non-carcinogenic risk $AT=ED \times EF$

The product of ED and EF is equal to that of the AT, thus simplifying equation to:

$$CDI = C \times IR / BW$$

Typical exposure parameters used in this study to calculate CDI are presented in Table 1 below and were drawn from USEPA (2011) exposure-factor recommendations and demographic exposure assumptions.

Table 1: Population groups and exposure parameters

Parameter	Adults	Children
1. Water daily intake rate assumed value	2 L/day	1 L/day
2. Body weight actual assumed value	70 kg	15 kg
3. Exposure duration (ED) actual assumed value	30 years	6 years
4. EF (exposure frequency (days/year) actual assumed value	365	365
5. Fluoride concentration	Measured value	Measured value
6. Reference dose for daily intake of fluoride	0.06 mg/kg/day	0.06 mg/kg/day

2.4.2 Non-carcinogenic health-risk assessment

The potential non-carcinogenic health risk associated with fluoride exposure was quantified using the Hazard Quotient (HQ), calculated as:

$$HQ = CDI / RfD$$

Where: HQ is hazard quotient (unitless), RfD is the oral reference dose for fluoride (0.06 mg/kg/day) and

CDI represents chronic daily fluoride intake.

An $HQ \leq 1$ was interpreted as indicating no appreciable non-carcinogenic risk, whereas $HQ > 1$ indicated potential health risk requiring attention, consistent with the USEPA risk-assessment framework.

2.5 Data statistical analysis

Descriptive statistics including the mean, maximum, minimum and range were computed to summarize fluoride concentration in sampled boreholes and the results interpreted by comparing fluoride concentrations with the World Health Organization drinking water guideline value of 1.5 mg/l to characterize contamination level. The statistical analysis was conducted using Microsoft excel. The mean values were taken to represent average fluoride concentration in each borehole. CDI and HQ were calculated separately for children and adults using the USEPA (2011) exposure equations. CDI calculated values were evaluated against the USEPA oral reference dose (RfD) of 0.06 mg/kg/day, where CDI values ≤ 0.06 mg/kg/day were considered to be within or below the health based reference dose, whereas CDI values > 0.06 mg/kg/day indicated intake above the reference dose and potential non-carcinogenic health concern. The potential non-carcinogenic risk was further evaluated using the Hazard Quotient ([HQ) index, with $HQ > 1$ indicating potential non-carcinogenic health risks and HQ below 1.0 safe exposure level.

2.6 Validity of the results

The validity of the results was enhanced through standardized groundwater sampling and fluoride analysis using calibrated equipment, consistent standard laboratory analytical procedures, and USEPA-based methods for exposure and non-carcinogenic health-risk assessment.

2.8 Ethical considerations

Ethical approval and permission to collect groundwater samples from Olbanita wellfield boreholes were obtained from NAWASSCO before fieldwork. Water sampling was conducted using standard procedures without disrupting water supply or damaging infrastructure. The study did not involve human participants; therefore, informed consent was not required. All data were handled confidentially, and the results were reported objectively and in accordance with accepted scientific and research integrity standards.

3.0 RESULTS

3.1 Fluoride Concentration in groundwater

Analytical results are presented in Table 2 below where mean fluoride concentrations ranged from 1.50 to 11.60 mg/l, indicating considerable spatial variability among the six sampled boreholes. Borehole BB7 recorded the lowest mean fluoride concentration of 1.50 mg/l, while the highest concentration occurred at Bb5 (11.60 mg/l). In between, mean fluoride concentrations of 6.28, 5.67, 5.67 and 7.33 mg/l were recorded at boreholes BB2, BB3, BB4 and BB6 respectively. Five of the six boreholes recorded fluoride concentrations substantially above the WHO (2017) recommended guideline value of 1.5 mg/l for drinking water.

Table 2: Groundwater fluoride concentration in mg/l (dry season – January to March)

Borehole ID	Minimum	Maximum	Mean	WHO guideline Compliance (mg/L)
BB2	5.23	7.00	6.2767	1.5
BB3	5.00	6.00	5.6667	
BB4	5.00	6.00	5.6667	
BB5	8.50	14.00	11.6000	
BB6	5.50	10.00	7.3333	
BB7	0.50	3.00	1.5000	

3.2 Calculated Chronic Daily Intake (CDI) and Hazard Quotient (HQ) for adults and children

Results are presented in Table 3 and 4 below.

Table 3: CDI and Hazard Quotient Values for adults

Borehole ID	CDI	HQ
BB2	0.1793	2.99
BB3	0.1619	2.70
BB4	0.1619	2.70
BB5	0.3314	5.52
BB6	0.2095	3.49
BB7	0.0429	0.71

Table 4: Calculated CDI and Hazard Quotient Values for children

Borehole ID	CDI	HQ
BB2	0.4184	6.97
BB3	0.3778	6.30
BB4	0.3778	6.30
BB5	0.7733	12.89
BB6	0.4889	8.15
BB7	0.1000	1.67

3.2.1 Chronic Daily Intake of fluoride

Results of calculated Chronic Daily Intake (CDI) of fluoride from groundwater water consumption indicate proportional increase to fluoride concentration in each borehole as above (Table 2, 3 and 4) and exposure variations among children and adults. CDI values ranged from 0.043 to 0.331 mg/kg/day for adults and 0.100 to 0.773 mg/kg/day for children, often exceeding the health based reference dose of 0.06 mg/kg/day. Across all boreholes, children exhibit higher daily fluoride intake than adults on a body adjusted weight as assumed in this study in Table 1. The maximum estimated CDI in children was approximately 2.3 times the maximum adult CDI. The highest CDI values for both population groups were associated with Borehole BB5, whereas the lowest values were observed at Borehole BB7.

3.2.2 Non-carcinogenic health risk

The Hazard Quotient (HQ) values calculated for adults and children are presented in Table 3 and 4. The estimated HQ differed substantially among boreholes and between exposure groups. HQ values ranged from 0.71 to 5.52 for adults and 1.67 to 12.89 for children. Borehole Bb7 was the only source with an adult HQ below the acceptable threshold (HQ = 0.71). For children, HQ values exceeded the acceptable limit of one unit in all sampled boreholes, indicating that all groundwater sources pose potential non-carcinogenic health risks to children. Adults exhibited HQ values greater than one unit in five of the six boreholes, indicating potential non-carcinogenic health risks associated with long-term consumption of groundwater from these sources. The highest HQ values occurred at BB5, the borehole with the highest fluoride concentration, whereas the lowest values occurred at BB7. The results generally show that the spatial pattern of HQ values across the six boreholes was consistent with the observed fluoride concentration and CDI patterns.

4.0 DISCUSSION

4.1 Groundwater fluoride concentration

The results showed considerable spatial variation in groundwater fluoride concentrations across the Olbanita boreholes, with mean concentrations ranging from 1.50 to 11.60 mg/l. Except for one borehole, all sampled groundwater sources exceeded the World Health Organization (WHO) guideline value of 1.5 mg/l for drinking water, indicating widespread fluoride contamination. The observed results are consistent with investigations of previous studies conducted within Nakuru County and the broader Kenyan Rift Valley where groundwater fluoride concentrations have been reported to frequently exceed recommended drinking water standards due to volcanic geology and complex hydrogeochemical processes characteristic of the East African Rift System (Keli, 2022; Mwiathi et al., 2022; Gevera & Mouri, 2018; Gevera et al., 2019; Olaka et al., 2016).

Gevera (2017) and Olaka et al. (2016) reported fluoride concentrations up to 72mg/l and 74.98 mg/l in Nakuru region, with most boreholes exceeding the (WHO, 2017) recommended limit of 1.5 mg/l for drinking water. Mean fluoride concentration values observed in the present study is lower but exceeded recommended guideline value confirming that elevated fluoride remains a significant groundwater-quality issue in Nakuru region. Comparable groundwater fluoride enrichment has also been documented globally, highlighting that geogenic contamination remains a major challenge in volcanic terrains worldwide (Ali et al., 2026; Ashong et al., 2024; Shube et al., 2023; Solanki et al., 2022; Malago et al., 2017; Rango et al., 2012). The spatial variability of fluoride concentration observed among the boreholes is also consistent with Gevera and Mouri (2018) and Sosi et al. (2020) observations that fluoride concentrations in Nakuru groundwater system are not simply controlled by aquifer type but vary according to location.

4.2 Estimated Chronic Daily Intake (CDI) of fluoride

Results of calculated CDI demonstrated substantial exposure to fluoride through consumption of borehole water, where boreholes with higher mean fluoride concentrations correspondingly produced higher values. Estimated exposure levels varied according to population characteristics and showed clear differences between adults and children. The estimated adult CDI ranged from 0.0429 to 0.3314 mg/kg/day and 0.1000 to 0.7733 mg/kg/day for children, exceeding the USEPA chronic oral reference dose of 0.06 mg/kg/day. The exceedance indicated a potential non-carcinogenic health concern associated with chronic fluoride exposure among adults and children in all the boreholes. Children experienced substantially higher fluoride exposure than adults despite consuming less quantities of water (Tables 3 and 4). The difference was primarily attributed to children's lower body weight and relatively higher daily water consumption per unit body mass, resulting in greater fluoride intake as reported in numerous health risk assessments worldwide, including studies by Ashong et al. (2024). He demonstrated that children consistently experience higher fluoride exposure than adults in groundwater-dependent communities. Similar observations have been reported in previous studies on health risk assessment by groundwater

geogenic fluoride ingestion in shallow aquifers of the Kenyan Rift Valley and other fluoride-endemic regions, where children experienced the highest chronic health threats (Abera et al., 2025; Mwiathi et al., 2022; Solanki et al., 2022).

4.3 Non-carcinogenic health risk analysis

The Hazard Quotient (HQ) values obtained in this study indicated differences in potential non-carcinogenic health risks among population groups, particularly among children. HQ values ranged from 1.67 to 12.89 for children, while values between 0.71 and 5.52 were recorded for adults, often exceeding USEPA limit ($HQ > 1$) (Table 3 and 4). All boreholes presented serious health threats to children under the exposure assumptions used in this study, confirming their greater vulnerability to fluoride related risks. The adult HQ range also indicated that fluoride exposure estimates exceeded one unit in five boreholes, demonstrating that potential risk was not restricted to children. The findings of higher HQ values patterns in children are consistent with Mwiathi et al. (2022), who reported greater fluoride-related risks among children than adults in shallow aquifers of Kenya Rift Valley, with high fluoride concentrations attributed mainly to geogenic processes associated with volcanic rocks. Similarly, global literature review of exposure risk studies continue to identify children as most vulnerable group because of their lower body weight and higher intake rates relative to their size (Kitemangu et al., 2026; Ashong et al., 2024; Solanki et al., 2022; Ullah et al. (2022).

4.4 Study strengths and limitations

4.4.1 Limitations

First, the investigations involved six Olbanita boreholes, so the results should not automatically be extrapolated to the entire broader Olobanita groundwater system. Second, CDI and HQ are dependent on exposure assumptions applied in this study (Table 1), hence represent estimates rather than direct measurements of individual fluoride intake. Third, the assessment focused on groundwater consumption pathway and did not analyze fluoride intake scenarios from food, beverages or dental products intake, although WHO recognizes these as additional contributors to total fluoride exposure. Fourth, the health-risk assessment was based on modeled exposure and reference-dose comparison rather than clinical examination of the exposed population. Therefore, the HQ findings indicate potential risk rather than confirmed disease

4.4.2 Strength of the study

The study notable strength included standardized fluoride analysis, integration of concentration with exposure assessment and non-carcinogenic health-risk for both adults and children using the USEPA-based risk framework to provide location-specific evidence to support groundwater quality monitoring and public-health protection.

4.4.3 Uncertainty or sensitivity analysis

Potential uncertainty in the risk estimates was acknowledged due to variations in measured fluoride concentration and assumed exposure parameters adopted from USEPA framework given in Table 1. Sensitivity analysis assessed the influence of key exposure parameters on estimated CDI and HQ.

6.0 CONCLUSION

This study assessed groundwater fluoride contamination and potential non-carcinogenic health risks through water consumption from Olbanita wellfield boreholes, Nakuru County, Kenya. The findings confirmed spatial variable elevated fluoride concentrations associated with the regions volcanic and hydrogeological characteristics. The estimated Chronic Daily Intake (CDI) and Hazard Quotient (HQ) values demonstrated significant exposure differences between adults and children and potential health risks. Higher CDI and HQ values observed among children compared to adults indicated their greater potential vulnerability to fluoride exposure and related potential non-carcinogenic health risks. Overall, the findings demonstrate that elevated groundwater fluoride in the Olbanita boreholes presents a significant water quality and public health problem where consumption remain an important pathway for fluoride exposure. The study contributes to theory, practice and policy by integrating fluoride exposure and health risk assessment with evidence based groundwater management and safe water policy. Therefore, regular source-specific monitoring of groundwater fluoride, adoption of alternative safe drinking supplies, integration of community health education and fluoride removal interventions are recommended.

Data availability statement

The data supporting the findings of this study are available from me as the corresponding author upon reasonable request.

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